

Pearl

Three-Phase Fe-C Phase-Field Solver

Physics, input, and execution guide for the serial and MPI programs

Purpose

This manual is written for the researcher who inherits the project. It connects each numbered code block to the underlying materials model, identifies every input group, and gives complete serial, parallel, output, and restart procedures.

System	Fields	Numerical form
Fe-C: ferrite + cementite + austenite	$\phi_\alpha, \phi_\theta, \phi_\gamma, C, \mu$	2D finite differences; explicit phase step; implicit carbon diffusion

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How to use this manual

The section numbers below are printed as comments in both `pearl_serial.cpp` and `pearl_parallel.cpp`. The scientific kernels are common to both files; Section 1.2 selects single-process or distributed execution. A new user can read Sections 2, 4, 5, and 7 for the working model, Section 8 for input preparation, and Sections 9-10 for operating the solver.

Code section	Subject	Primary use
1	Program identity and execution mode	Select serial or MPI build
2	Thermodynamic model	Understand phase free energies, grand potentials, and closure
3	Conservative carbon diffusion	Understand the implicit chemical-potential solve
4	Interfacial free energy	Understand interface and triple-junction energetics
5	Phase-field kinetics	Understand phase evolution and interface profiles
6	Computational domain and MPI services	Understand decomposition, halos, and boundary conditions
7	Coupled evolution and time integration	Follow one accepted physical time step
8	Input processing and initialization	Prepare a complete input file
9	Output, checkpoints, and moving window	Interpret files and continue a run
10	Application control and execution	Build, launch, monitor, and stop

Model at a glance

The model is isothermal and represents pearlite growth from a prescribed α/θ product geometry into γ . Thermodynamics are expressed with phase-specific parabolic free energies and a common carbon chemical potential. Interfacial energy is isotropic with one pairwise σ and a triple-phase term. Carbon transport follows the chemical-potential gradient. The x direction is periodic and the y boundaries are zero flux.

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Section 1 - Program identity and execution mode

Section 1 establishes checkpoint identity and selects the execution path. The serial and parallel translation units contain the same physical model, input reader, time integrator, and output routines.

Section 1.1 - Checkpoint model identifier

PEARL_SOURCE_DIGEST is included in the physics signature written to every field frame and checkpoint. During restart, the saved signature is compared with the signature produced from the new input. This prevents a checkpoint from being paired with a different grid or physical parameter set.

Section 1.2 - Execution-mode selection

In `pearl_serial.cpp`, PEARL_MPI remains undefined and the domain contains one process. In `pearl_parallel.cpp`, the macro is defined before the shared implementation, enabling MPI initialization, Cartesian decomposition, halo exchange, and collective reductions.

Executable	Domain layout	Recommended use
pearl_serial	1 x 1 process grid	Short calculations, input preparation, and local checks
pearl_parallel	px x py Cartesian process grid	Production calculations on an allocated compute node

Section 2 - Thermodynamic model

The local thermodynamic state is defined by the three phase fractions, the total carbon atomic fraction, and a common carbon chemical potential. The phases are α ferrite, θ cementite, and γ austenite.

Section 2.1 - Phase indexing and field types

Index	Code name	Metallurgical phase	Field
0	Alpha	Ferrite, α	ϕ_α
1	Theta	Cementite, θ	ϕ_θ
2	Gamma	Austenite, γ	ϕ_γ

The phase fractions satisfy $\phi_\alpha + \phi_\theta + \phi_\gamma = 1$ and $0 \leq \phi_p \leq 1$ at every cell. Carbon concentration c is an atomic fraction.

Section 2.2 - Parabolic phase free energies and grand potentials

Each phase is represented by a convex molar free-energy parabola at the run temperature. This local approximation provides an analytical chemical-potential relation, susceptibility, and Legendre transform.

$$f_p(c_p) = A_p c_p^2 + B_p c_p + C_p$$

$$\mu = 2A_p c_p + B_p \quad c_p(\mu) = (\mu - B_p)/(2A_p) \quad \chi_p = 1/(2A_p)$$

$$\psi_p(\mu) = f_p(c_p) - \mu c_p = C_p - (\mu - B_p)^2/(4A_p)$$

Here ψ_p is the phase grand-potential density on a molar basis. The input diffusivity D_p is stored with each phase parabola.

Section 2.3 - Gibbs-simplex constraint and phase interpolation

Thermodynamic properties are interpolated with a three-phase cubic function that preserves partition of unity on the Gibbs simplex:

$$h_p(\phi) = \phi_p^2(3 - 2\phi_p) + 2\phi_\alpha\phi_\theta\phi_\gamma, \quad \sum_p h_p = 1$$

The code evaluates both h_p and $\partial h_p / \partial \phi_a$. The derivatives enter the chemical contribution to the phase-field driving force.

Section 2.4 - Mixture composition, susceptibility, and transport mobility

At a multiphase cell, the KKS-type local closure is linear in μ because every phase free energy is parabolic:

$$\begin{aligned} \chi(\phi) &= \sum_p h_p \chi_p & c_0(\phi) &= -\sum_p h_p \chi_p B_p \\ c(\phi, \mu) &= c_0(\phi) + \chi(\phi)\mu & K(\phi) &= \sum_p \phi_p D_p \chi_p \end{aligned}$$

The thermodynamic interpolation uses h_p , whereas transport uses the phase fractions ϕ_p . K has the mobility required for chemical-potential-driven carbon flux.

Section 2.5 - Chemical driving force and chemical free energy

The phase-field chemical force is the derivative of the interpolated grand potential with respect to each phase fraction. The common molar volume converts molar grand potential to volumetric driving-force density.

$$\partial \Omega_{\text{chem}} / \partial \phi_a = (1/V_m) \sum_p \psi_p(\mu) \partial h_p / \partial \phi_a$$

The helper routines also reconstruct μ from c and ϕ and evaluate the phase-weighted chemical free energy at local equilibrium.

Section 2.6 - Temperature-dependent grand-potential preparation

For `eutectoid_reference` input, the run is isothermal at $T = T_E - \Delta T$. The supplied reference parabolas are first transformed to grand-potential coefficients:

$$q_2 = -1/(4A), \quad q_1 = B/(2A), \quad q_0 = C - B^2/(4A)$$

$$q_k(T) = q_k(T_E) + (T - T_E)(dq_k/dT)$$

The updated q coefficients are converted back to A , B , and C before initialization. In `direct_at_temperature` mode, A , B , and C already correspond to the stated run temperature and are used directly.

Section 3 - Conservative carbon diffusion

Carbon transport is formulated through gradients in chemical potential. Section 3 provides the whole-domain form of the diffusion operator; Section 7.5 applies the same formulation through the serial/MPI domain abstraction used by the executable.

Section 3.1 - Uniform grid and scalar/phase-field storage

The physical domain contains $n_x \times n_y$ cell-centered values with spacings dx and dy . Scalar fields store c or μ ; the phase field stores three components per cell.

Section 3.2 - Unique cell faces and arithmetic face mobility

Each unique east and north face is assembled once. The face mobility is the arithmetic mean of the adjacent cell mobilities. Periodic x faces connect the last and first columns; boundary faces normal to y are omitted, producing zero carbon flux.

$$J_c = -K(\phi)\nabla\mu \quad \text{and} \quad \partial c/\partial t = \nabla \cdot [K(\phi)\nabla\mu]$$

Section 3.3 - Backward-Euler operator and preconditioned CG solver

After the phase field has been updated, χ , c_0 , and K are fixed for the diffusion substep. Backward Euler gives a symmetric positive-definite linear system for μ^{n+1} :

$$[\chi - \Delta t \nabla \cdot (K \nabla)] \mu^{n+1} = c^n - c_0$$

A diagonally preconditioned conjugate-gradient method solves the system. The tolerance is $\text{cg_atol} + \text{cg_rtol} ||\text{rhs}||_2$.

Section 3.4 - Conservative carbon update and thermodynamic closure

The accepted face fluxes are added to one cell and subtracted from its neighbor. This pairwise update conserves total carbon to round-off. The code then reconstructs μ from the updated composition and verifies the local closure $c = c_0 + \chi\mu$.

Section 4 - Interfacial free energy

The phase-field contribution combines an isotropic pairwise gradient term, a multi-obstacle potential, and a triple-phase term. With equal pairwise interfacial energy σ , the implemented functional is:

$$F_{\text{int}} = \int_{\Omega} \{ \epsilon \sigma \sum_{a<b} |\phi_a \nabla \phi_b - \phi_b \nabla \phi_a|^2 + [16\sigma/(\pi^2 \epsilon)] \sum_{a<b} \phi_a \phi_b + (g_3/\epsilon) \phi_\alpha \phi_\theta \phi_\gamma \} dV$$

Section 4.1 - Interface width, pair energy, and triple-phase coefficient

Input	Symbol	Physical role
epsilon	ϵ	Diffuse-interface length scale
sigma	σ	Common α - θ , α - γ , and θ - γ interfacial energy
triple	g_3	Controls energetic occupancy of three-phase junction regions

Section 4.2 - Local obstacle and triple-phase contributions

The obstacle contribution penalizes coexistence of each phase pair across a finite interface. The triple product contributes only where all three phase fractions are nonzero and therefore acts at diffuse three-phase junctions.

Section 4.3 - Pairwise gradient contribution on cell faces

The antisymmetric pair measure $\phi_a \nabla \phi_b - \phi_b \nabla \phi_a$ is evaluated on grid faces. Its discrete derivative is accumulated into both neighboring cells and both participating phases, giving a consistent capillary force.

Section 4.4 - Projection of phase fractions onto the Gibbs simplex

Every trial phase vector is projected onto the simplex $\phi_p \geq 0$ and $\sum \phi_p = 1$. This maintains physically admissible phase fractions without modifying the carbon field.

Section 5 - Phase-field kinetics

Section 5.1 - Pairwise relaxation coefficients and local kinetic response

The three kinetic inputs are $\tau_{\alpha\theta}$, $\tau_{\alpha\gamma}$, and $\tau_{\theta\gamma}$. In a diffuse multiphase cell, the local relaxation coefficient is interpolated with pair products:

$$\tau(\varphi) = [\varphi_{\alpha}\varphi_{\theta}\tau_{\alpha\theta} + \varphi_{\alpha}\varphi_{\gamma}\tau_{\alpha\gamma} + \varphi_{\theta}\varphi_{\gamma}\tau_{\theta\gamma}] / [\varphi_{\alpha}\varphi_{\theta} + \varphi_{\alpha}\varphi_{\gamma} + \varphi_{\theta}\varphi_{\gamma}]$$

Section 5.2 - Explicit constrained phase-field update

The capillary and chemical derivatives are summed to form $G_a = \delta F / \delta \varphi_a$. The mean force is removed to enforce the phase-sum constraint, followed by a forward-Euler update and simplex projection:

$$\varphi_a^{n+1} = P_{\text{simplex}} \{ \varphi_a^n - [\Delta t / (\tau \varepsilon)] [G_a - (1/3) \sum_b G_b] \}$$

The largest phase-fraction change over the domain is returned to the adaptive step controller.

Section 5.3 - Compact diffuse-interface profile for initialization

The analytical initialization profile equals one on the negative-distance side, zero on the positive-distance side, and varies smoothly over a compact interval:

$$\varphi(d) = 0.5[1 - \sin(4d/(\pi\varepsilon))] \text{ for } |d| < \pi^2\varepsilon/8$$

This profile is used by the lamellar, flat, and circular initialization modes.

Section 6 - Computational domain and MPI services

Section 6.1 - Local arrays with one-cell ghost layers

Each process stores its local interior cells plus one ghost cell on every side. Scalar arrays have one component; the phase array has three. All physical calculations are performed on local interior cells.

Section 6.2 - Cartesian domain decomposition and local grid extents

The global grid is divided into p_x blocks in x and p_y blocks in y . The MPI rank count must equal $p_x \times p_y$. Remainder cells are assigned to the lower coordinate blocks, so exact divisibility is not required.

Example	Ranks	Command-line layout	Interpretation
Serial	1	$p_x=1, p_y=1$	Entire grid on one process
MPI	16	$p_x=4, p_y=4$	Four blocks across x and four across y
MPI	8	$p_x=4, p_y=2$	Useful when the domain is wider than it is tall

Section 6.3 - Global reductions and synchronization

Global sums, maxima, minima, logical agreement, and barriers are implemented as MPI collectives in the parallel executable and as direct operations in the serial executable. They support global mass balance, norms, energy, bounds, and synchronized file operations.

Section 6.4 - Halo exchange and physical boundary conditions

East-west halos are exchanged through the periodic Cartesian direction. At the physical south and north boundaries, the nearest interior value is copied into the ghost cell, imposing zero normal derivative. The same boundary treatment is applied to phase fractions, carbon, chemical potential, and mobility.

Section 6.5 - Global assembly and redistribution for I/O and shifting

For field output, checkpoints, and moving-window remapping, local interiors are gathered and ordered as one global array on rank 0. Restart data and remapped fields are broadcast and each process extracts its new local block. Ghost values are regenerated after loading.

Section 7 - Coupled evolution and time integration

One physical time step consists of an explicit phase update followed by an implicit diffusion update. The code accepts the trial state only when phase bounds, thermodynamic closure, carbon conservation, and the free-energy criterion are satisfied.

Section 7.1 - Evolving state, solver controls, and step diagnostics

The evolving state contains ϕ , c , μ , physical time, proposed next time step, global shift distance, carbon ledger, accepted-step count, shift count, and rejected-step count. Solver controls specify phase-change, conjugate-gradient, closure, energy, and time-step limits.

Section 7.2 - Thermodynamic closure, total carbon, and free-energy measures

The runtime reconstructs μ from c and ϕ when needed, evaluates the global carbon inventory, and evaluates a reduced free energy relative to the initial reference chemical potential μ_{ref} . The reduced energy contains chemical, gradient, obstacle, and triple-phase contributions and is used as the step-acceptance Lyapunov measure.

Section 7.3 - Capillarity-controlled preprocessing of sharp interfaces

For `initialization = sharp_smooth`, exact phase labels are relaxed using the interfacial free energy alone. Backtracking selects energy-decreasing pseudo-steps. Carbon and μ are assigned after smoothing, and physical time starts at zero.

Section 7.4 - Domain-aware phase-field proposal

The phase proposal evaluates chemical and capillary forces with local cells and exchanged halos. A common ψ_Y contribution is subtracted before projection because it does not alter constrained phase evolution. The global maximum $|\Delta\phi|$ is returned for acceptance testing.

Section 7.5 - Domain-aware implicit diffusion solve

The executable solves for $u = \mu - \mu_{\text{ref}}$. Centering the unknown does not change its gradients and improves numerical scaling. The parallel conjugate-gradient algorithm uses halo exchange for every matrix application and global dot products for residuals. After convergence, carbon is updated from conservative face fluxes and μ is reconstructed from local closure.

Section 7.6 - Step acceptance, conservation checks, and time-step adaptation

The acceptance sequence is:

Stage	Accepted when
1. Phase proposal	The global maximum phase change is finite and does not exceed max_phase_change
2. Diffusion	CG converges and the closure residual is within closure_tol
3. State bounds	Phase simplex, total carbon, phase compositions, and chemical potential are finite and admissible
4. Conservation	Total carbon change is within the global mass tolerance
5. Free energy	The reduced free energy does not rise beyond the numerical tolerance

An accepted step advances time and proposes $\min(dt_max, 1.1 \ dt)$ for the next step. A rejected trial halves Δt and is attempted again. The run stops with an error if Δt falls below dt_min or if 24 retries are exhausted.

Working time-step sequence

old state → exchange phase halos → phase proposal → implicit carbon diffusion → state and conservation checks → free-energy check → accept and advance, or halve Δt and retry

Section 8 - Input processing and initialization

Section 8.1 - Configuration data model and checkpoint signature

The Config object holds the grid, thermodynamics, interfacial parameters, kinetics, time controls, initialization, and moving-window settings. These physical settings are serialized into the checkpoint signature used during restart.

Section 8.2 - Key-value parser

An input file contains one key = value pair per line. Text after # is ignored. Duplicate keys, unknown keys, missing required values, nonfinite values, and values outside the allowed bounds stop before initialization.

```
# Example syntax
thermo_mode = eutectoid_reference
T_eutectoid = 999.76
undercooling = 10.0
nx = 137
ny = 250
dx = 2.79e-9
```

Section 8.3 - Physical parameters, numerical controls, and validation

Thermodynamic modes

Mode	Required thermodynamic entries
eutectoid_reference	T_eutectoid, undercooling, max_undercooling, temperature_law=linear_gp; Aeq, Beq, Ceq, dq2_dT, dq1_dT, dq0_dT, and D for alpha, theta, gamma
direct_at_temperature	temperature; A, B, C, and D for alpha, theta, gamma

Geometry, interface, and kinetics

Key	Meaning	Requirement or default
nx, ny	Global numbers of cells	Required
dx	x cell spacing	Required
dy	y cell spacing	dx
Vm	Common molar volume	Required
epsilon	Diffuse-interface length	Required
sigma	Common pairwise interfacial energy	Required
triple	Triple-phase energy coefficient	Required
tau_at, tau_ag, tau_tg	Pairwise kinetic relaxation coefficients	Required

Time and linear-solver controls

Key	Meaning	Requirement or default
dt	Initial physical time step	Required
dt_max	Maximum time step	dt
dt_min	Minimum retried time step	dt / 1048576
end_time	Simulation horizon	Required
output_dt	Physical time between frames	Required
max_steps	Absolute accepted-step ceiling	100000000
max_phase_change	Maximum $ \Delta\phi $ per accepted step	0.04
cg_rtol, cg_atol	Relative and absolute CG tolerances	1e-11, 1e-14
cg_max	Maximum CG iterations	2000
closure_tol	Maximum c- μ closure residual	1e-10
energy_atol	Absolute free-energy acceptance tolerance	Computed from σ and domain size
mode	coupled or phase_only	coupled

Moving-window controls

Key	Meaning	Default
moving_window	0 disables; 1 enables y translation	0
shift_trigger	Front position at which rows shift	0.65 ny dy
shift_target	Front position after the shift	0.35 ny dy
shift_gamma_carbon	Carbon fraction in inserted gamma rows	Initial gamma-state composition

Section 8.4 - Initial phase geometry

Initialization	Phase geometry	Associated inputs
lamella	Diffuse α/θ product below seed_height; γ above	seed_height, theta_columns
sharp_smooth	Sharp α/θ product and γ followed by interfacial relaxation	seed_height, theta_columns, smooth_steps, smooth_min_steps, smooth_energy_tol, smooth_max_phase_change
flat	Planar interface normal to y	seed_height, phase_a, phase_b
circle	Circular phase_a region inside phase_b	radius, phase_a, phase_b

The default geometry values are seed_height = $n_y \, dy / 4$, theta_columns = floor($n_x/2$), radius = min($n_x \, dx$, $n_y \, dy$)/4, phase_a = 0, and phase_b = 2.

Section 8.5 - Initial composition and chemical-potential closure

Exactly one composition specification is required:

Input	Result
initial_mu	Uses the stated uniform μ and evaluates local c from the phase mixture
gamma_carbon	Evaluates μ from the γ parabola at the stated c_γ , then applies that uniform μ throughout the domain
mean_carbon	Solves one uniform μ that gives the requested spatial mean carbon fraction for the initialized phase geometry

Unit convention

Lengths are in metres, physical times in seconds, temperatures in kelvin, carbon values in atomic fraction, phase free energies and μ in J/mol, D in m^2/s , σ in J/ m^2 , and V_m in m^3/mol .

Section 9 - Output, checkpoints, and moving window

Section 9.1 - HDF5 resource management, attributes, and datasets

HDF5 objects are opened and closed through scoped wrappers. Field datasets are written as 64-bit floating-point arrays with chunking and Fletcher32 checksums. Temporary .part files are renamed only after a successful write.

Section 9.2 - Field frames and XMF metadata

Every physical frame contains five two-dimensional datasets: alpha, theta, gamma, carbon, and mu. HDF5 attributes record time, next proposed Δt , step, spacing, temperature, moving-window displacement, carbon ledger, reference μ , and rejected-step count. The matching XMF file describes the grid for ParaView.

Section 9.3 - Checkpoint loading and state reconstruction

Rank 0 reads the global fields and metadata, verifies the file format and physics signature, and distributes local blocks. The restored state is checked for phase bounds, c - μ closure, and the carbon ledger before time integration resumes. A checkpoint can be repartitioned because it stores global fields without ghost layers.

Section 9.4 - Moving-window translation and carbon balance

At an output time, the front is located from the $\phi_\gamma = 0.5$ contour. If it exceeds shift_trigger, enough bottom rows are removed to return it near shift_target. Fresh γ with shift_gamma_carbon is inserted at the top, μ is closed again, and added/removed carbon is recorded.

Section 9.5 - Phase-only snapshots during interface preprocessing

The `sharp_smooth` path writes `init_sharp` and `init_smoothed` phase-only files. They allow the inherited user to inspect the prescribed geometry and the relaxed diffuse interfaces before physical evolution.

Section 9.6 - Evaluated thermodynamic parameter report

`thermodynamics.txt` records the run temperature, input mode, evaluated A/B/C and q coefficients, phase susceptibilities, diffusivities, initial composition mode, and kinetic coefficients. Use it as the concise record of the thermodynamic state actually passed to the solver.

Files in a run directory

File	Contents
Input.in	Copy of the input used for the run
thermodynamics.txt	Evaluated isothermal thermodynamic parameters
initialization_report.txt	Interface-smoothing and initial phase-fraction summary
init_sharp.h5/.xmf	Sharp phase geometry for <code>sharp_smooth</code>
init_smoothed.h5/.xmf	Relaxed phase geometry for <code>sharp_smooth</code>
frame_*.h5/.xmf	Time-ordered physical fields
diagnostics.csv	Global numerical and physical diagnostics at each frame
checkpoint.h5/.xmf	Final restart state
COMPLETED.txt	The requested <code>end_time</code> was reached
STOPPED.txt	A graceful stop or command-line step limit was reached

ParaView

- Open the XMF files rather than the HDF5 files; XMF supplies grid spacing, origin, time, and dataset links.
- Load the numbered frames as a file series to animate growth.
- Inspect alpha, theta, and gamma for morphology; carbon for partitioning; and mu for the diffusion potential.

Section 10 - Application control and execution

Section 10.1 - Command-line processing and MPI startup

```
pearl INPUT_FILE NEW_OUTPUT_DIR [--restart FILE.h5] [--px N --py N]
      [--max-steps N] [--thermo-only]
```

Argument	Meaning
INPUT_FILE	Path to the key-value input
NEW_OUTPUT_DIR	New directory created for all run products
--px N --py N	MPI process-grid dimensions; product must equal the MPI rank count
--restart FILE.h5	Load a compatible checkpoint
--max-steps N	Stop after N additional accepted steps and write a checkpoint
--thermo-only	Write Input.in and thermodynamics.txt, then exit without initialization or evolution

Section 10.2 - Domain creation, run directory, and initial state

The main program reads the input, constructs the serial or Cartesian MPI domain, creates the new output directory, and writes the evaluated thermodynamic report. It then either initializes the fields or loads a checkpoint. The output directory name must be new for every launch.

Section 10.3 - Run diagnostics and output schedule

The first frame is written at the initialized or restarted time. Subsequent steps are shortened when necessary to land on each output_dt boundary. The diagnostics table records mean carbon, carbon-ledger error, reduced energy, phase-sum error, field extrema, CG iterations, closure residual, rejected steps, shift distance, and wall time.

Section 10.4 - Snapshot assembly and field output

Each process contributes local extrema, norms, and fields. Global diagnostics are reduced across ranks; fields are gathered to rank 0 and written as one HDF5/XMF pair.

Section 10.5 - Time-marching, graceful stop, and final checkpoint

The loop advances until end_time, the input max_steps, a command-line --max-steps limit, or a stop signal. A graceful stop writes the current frame, checkpoint.h5, and STOPPED.txt. Reaching the time horizon writes COMPLETED.txt.

Build the serial executable

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build -j 4
```

Build serial and MPI executables

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release -DPEARL_ENABLE_MPI=ON
cmake --build build -j 4
```

Serial launch

```
./build/pearl_serial examples/fe_c_eutectoid.in run_serial
```

MPI launch: 16 ranks, 4 x 4 decomposition

```
mpirun -np 16 ./build/pearl_parallel \
  examples/fe_c_eutectoid.in run_parallel --px 4 --py 4
```

Open MPI/HPC-X launch

```
mpirun --bind-to none -mca coll_hcoll_enable 0 -np 16 \
  ./build/pearl_parallel examples/fe_c_eutectoid.in run_parallel \
  --px 4 --py 4
```

Slurm launch

```
#!/bin/bash
#SBATCH --nodes=1
#SBATCH --ntasks=16
#SBATCH --time=24:00:00
#SBATCH --partition=public
#SBATCH --job-name=pearl

srun ./build/pearl_parallel examples/fe_c_eutectoid.in run_parallel \
  --px 4 --py 4
```

Thermodynamic evaluation only

```
./build/pearl_serial examples/fe_c_eutectoid.in thermo_report --thermo-only
```

Controlled short run

```
./build/pearl_serial examples/fe_c_smoke.in smoke_run --max-steps 100
```

Restart

Prepare a compatible input with end_time later than the checkpoint time and choose a new output directory.

```
./build/pearl_serial continuation.in continuation_serial \
  --restart run_serial/checkpoint.h5

mpirun -np 16 ./build/pearl_parallel continuation.in continuation_parallel \
  --restart run_parallel/checkpoint.h5 --px 4 --py 4
```

Recommended operating sequence

- Compile and run examples/fe_c_smoke.in to confirm the environment and output path.
- Run --thermo-only after editing thermodynamic inputs and inspect thermodynamics.txt.
- Use --max-steps for a short field and diagnostics review.
- Open the XMF series and confirm the initial geometry, phase closure, carbon distribution, and growth direction.
- Submit the full serial or MPI calculation and retain checkpoint.h5 for continuation.

Quick reference

Task	Command or file
Short local run	./build/pearl_serial examples/fe_c_smoke.in smoke_run
16-rank run	mpirun -np 16 ./build/pearl_parallel input.in run --px 4 --py 4
Evaluate thermodynamics	--thermo-only, then read thermodynamics.txt
Stop cleanly	Ctrl-C or send SIGINT/SIGTERM
Continue	Use a new output directory and --restart previous/checkpoint.h5
Visualize	Open frame_*.xmf as a ParaView series
Monitor	Read diagnostics.csv

Handoff reading path

Start with the supplied Fe-C input and thermodynamics.txt. Then read code Sections 2, 4, 5, and 7 in order. Use Sections 8-10 when preparing and launching a new calculation.